

Homework #6 Unobserved Continuous Heterogeneity -- Rcpp

`Test_2.RData` contains panel data from a heterogeneous (“mixed”) logit model. Each choice set features two inside alternatives, “A” and “B,” and the outside good. The inside alternatives are vertically differentiated. At the same price, every consumer weakly prefers brand A to brand B. In line with vertical differentiation, A is priced above B. Prices vary across choice sets and consumers. Brand and price coefficients are heterogeneous in the population from which the panel data is sampled.

- 1) Analyze the marginal as well as the brand-conditional price distributions in the panel graphically and numerically.
- 2) Analyze the marginal distribution of choices and the prevalence of simulated consumers that always choose brand “A” (brand “B”).
- 3) Combining your insights from (1) and (2), what is your intuition about how informative the data are about consumers’ brand preferences and price sensitivities?
- 4) Fit a Bayesian homogenous logit model to this data ignoring consumer heterogeneity, e.g., using `rmnlIndepMetrop` in `bayesm`. Investigate the joint posterior of brand and price coefficients in this model graphically. What do you conclude about the joint empirical identification of the coefficients indexing this model?
- 5) Estimate i) a standard hierarchical model, ii) a model that sign-constrains the price coefficient to be negative, iii) a model that leaves the price coefficient unconstrained but constrains brand “A” coefficients to be weakly larger than brand “B” coefficients, and finally a model that constrains price coefficients to be negative and brand “A” coefficients to be weakly larger than brand “B” coefficients. Assess convergence for all models graphically and then compare converged posteriors, at all levels of the hierarchy numerically and graphically. How do the different models recover what you know about the data generating mechanism, i.e., that everybody prefers brand “A” to brand “B” and that everybody reacts negatively to price, of course?

BTW, the data generating preferences were simulated as follows:

```
library(MASS); nunits=1000
#Variance-covariance matrix of betastars
sigma = diag(c(0.3, .5, 1), nrow=3, ncol=3) #Specify variance
covariance matrix of normally distributed RV
#Expected values
avgbeta = c(-0.2, -2, 3) ###price, log delta brand A, brand B,
#Draw
betastar = mvrnorm(n=nunits, avgbeta, sigma) #draw true betas
from assumed distribution of heterogeneity
beta=betastar
#Price
beta[,1] = -exp(betastar[,1])
#Brand
beta[,2] = betastar[,3] + exp(betastar[,2])
```